

# GIRI-DRISHTI C-UAS GIMBAL PLATFORM

Itemized Bill of Materials (BoM) — High-Altitude Environmental Hardening Specifications

CONFIDENTIAL / DEFENSE SPEC

TRL-4 PRE-PROTOTYPE BASELINE

**Executive Cost & Sourcing Overview:** This document provides the formal itemized Bill of Materials (BoM) for the Giri-Drishti 2-axis direct-drive counter-UAS gimbal platform. Hardened for high-altitude operation (5,000 m ASL / 540 mbar / -55°C), the baseline platform replaces active resistive heating with passive athermalization and active observer control, achieving an **88% power savings**.

**Financial Summary & Currency Conversion:**

- **Total Prototype Budget:** ₹6,41,698 INR (\$7,685.00 USD)
- **Reference Exchange Rate:** 1 USD (\$) = ₹83.50 INR
- **Primary Cost Driver:** Optomechanical Yoke Chassis (SiC<sub>p</sub>/Al MMC) @ ₹1,54,475 INR (24.1% of total)

| Item                        | Subsystem Component   | Manufacturer / Part No.     | Material / Spec                          | Qty | Unit Cost (₹) | Total Cost (₹) | High-Altitude Hardening Feature                |
|-----------------------------|-----------------------|-----------------------------|------------------------------------------|-----|---------------|----------------|------------------------------------------------|
| 1.0                         | Gimbal Yoke Chassis   | Custom Machined             | SiC <sub>p</sub> /Al Composite (65% SiC) | 1   | ₹1,54,475     | ₹1,54,475      | Low CTE (6.5e-6/K), high stiffness at -55°C    |
| 2.1                         | Azimuth Drive Motor   | TQ-RoboDrive ILM-70x18      | Frameless PMSM Direct Drive              | 1   | ₹1,00,200     | ₹1,00,200      | Vacuum/low-pressure rated winding insulation   |
| 2.2                         | Elevation Drive Motor | TQ-RoboDrive ILM-50x14      | Frameless PMSM Direct Drive              | 1   | ₹79,325       | ₹79,325        | Zero-backlash direct coupling                  |
| 3.1                         | Azimuth Encoder       | RLS AksIM-2                 | 19-bit Absolute Magnetic Ring            | 1   | ₹56,780       | ₹56,780        | Non-contact, immune to condensation/ice        |
| 3.2                         | Elevation Encoder     | RLS AksIM-2                 | 19-bit Absolute Magnetic Ring            | 1   | ₹56,780       | ₹56,780        | High shock/vibration tolerance (MIL-STD-810H)  |
| 4.0                         | Flight Controller     | Custom STM32F767 / TI C2000 | 4-Layer Polyurethane Coated              | 1   | ₹35,070       | ₹35,070        | 1.48x IEC 60664-1 PCB trace gap (0.67mm)       |
| 5.0                         | Power Drivers         | GaN Systems GS66508B        | 650V GaN HEMT Bridges                    | 2   | ₹25,885       | ₹51,770        | 3.19x dielectric safety factor @ 540 mbar      |
| 6.0                         | Breathing Vent Valve  | Donaldson ePTFE Membrane    | IP67 Hydrophobic Vent                    | 2   | ₹3,758        | ₹7,515         | Continuous pressure equalization at 5,000m ASL |
| 7.0                         | Rotary Shaft Seals    | Trelleborg Turcon Variseal  | Fluorosilicone / PTFE Spring             | 2   | ₹7,098        | ₹14,195        | Rated to -65°C, low stiction grease            |
| 8.0                         | Precision Bearings    | Kaydon Reali-Slim           | 440C Stainless / MIL-PRF-23827           | 4   | ₹20,040       | ₹80,160        | Low-temp grease prevents sub-zero lockup       |
| 9.0                         | Thermal Insulation    | Aspen Aerogel Pyrogel XTE   | 3mm Flexible Aerogel Blanket             | 1   | ₹5,428        | ₹5,428         | k = 0.015 W/m-K thermal protection             |
| TOTAL PROTOTYPE SYSTEM COST |                       |                             |                                          | -   | -             | ₹6,41,698      | 88% power savings vs 320W active heater COTS   |